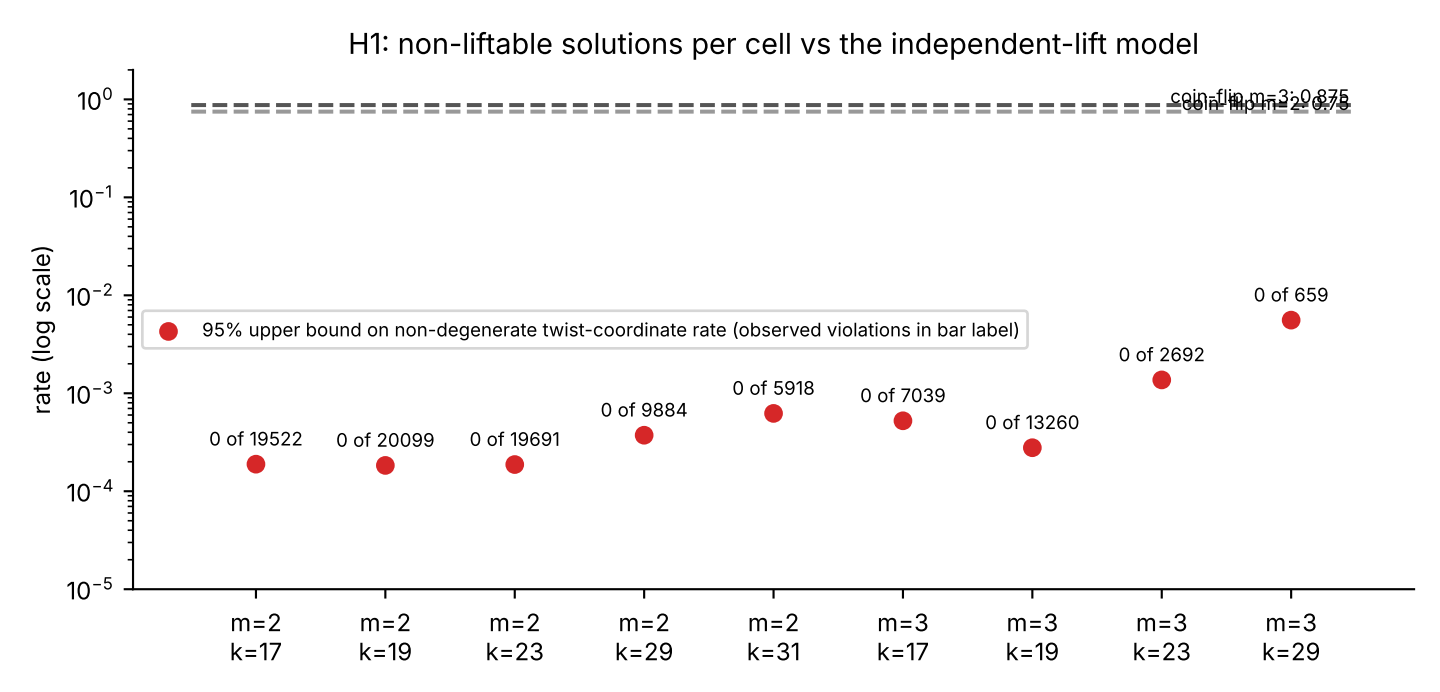
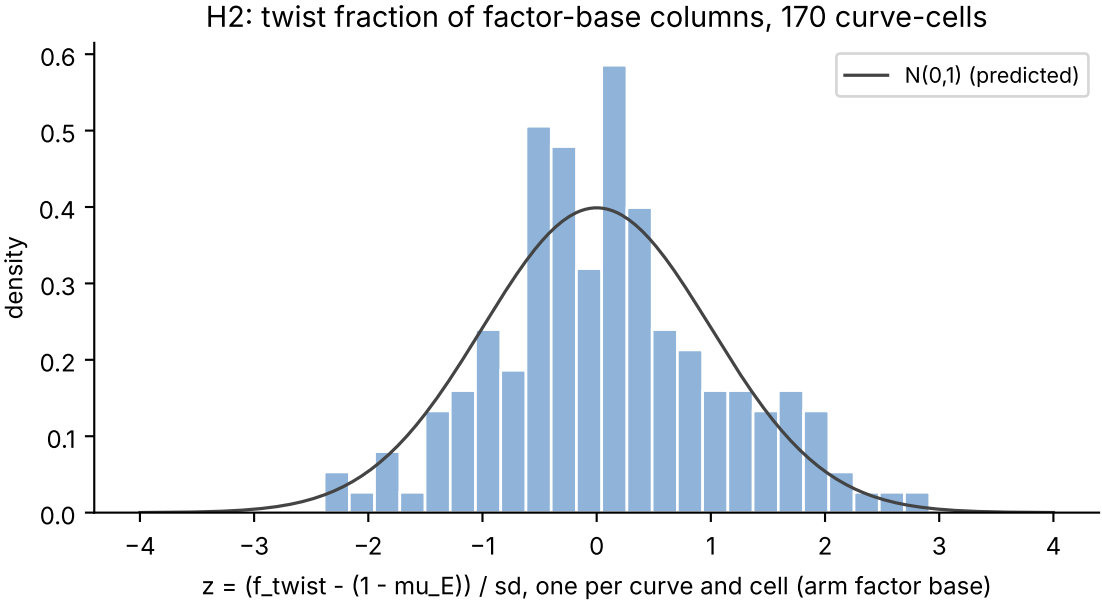
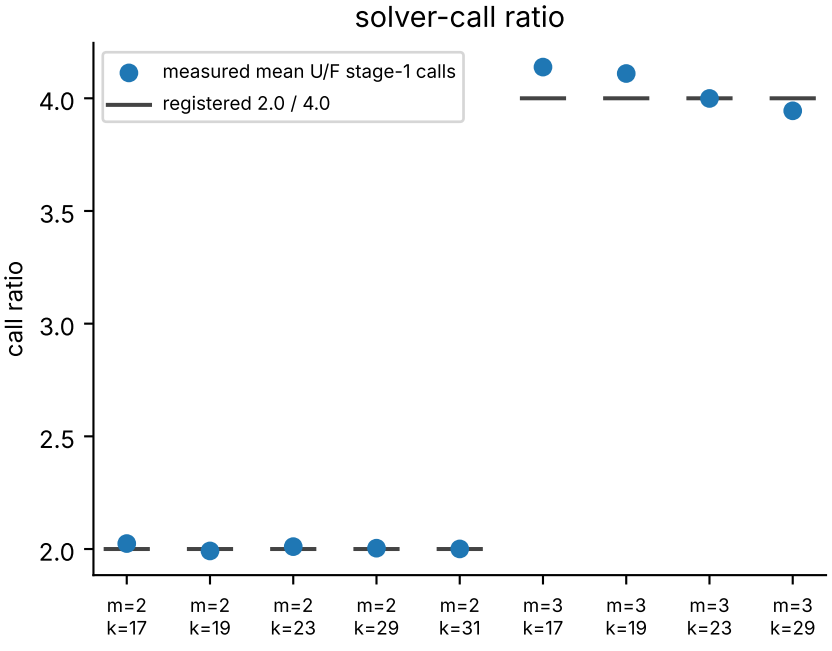
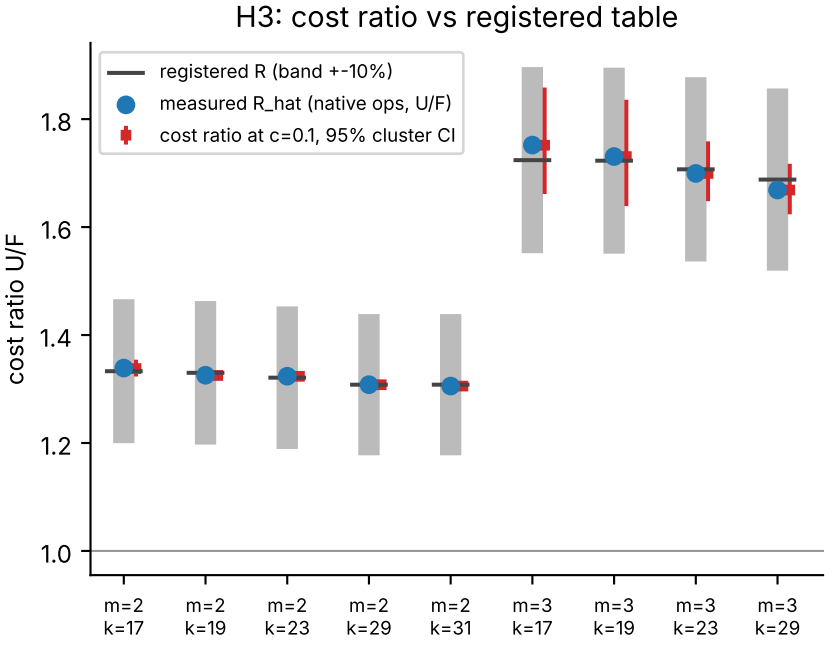
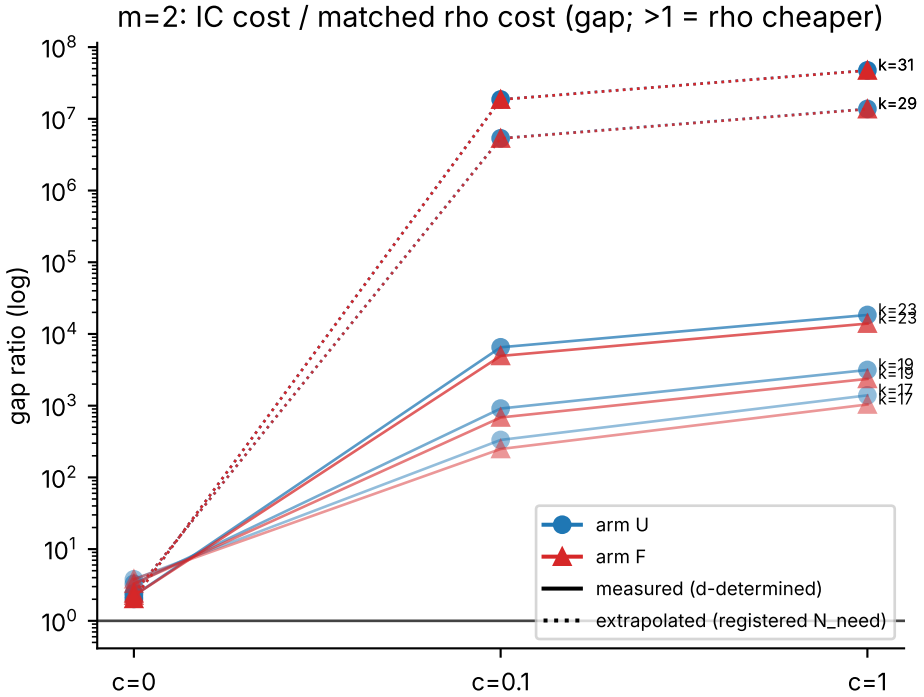
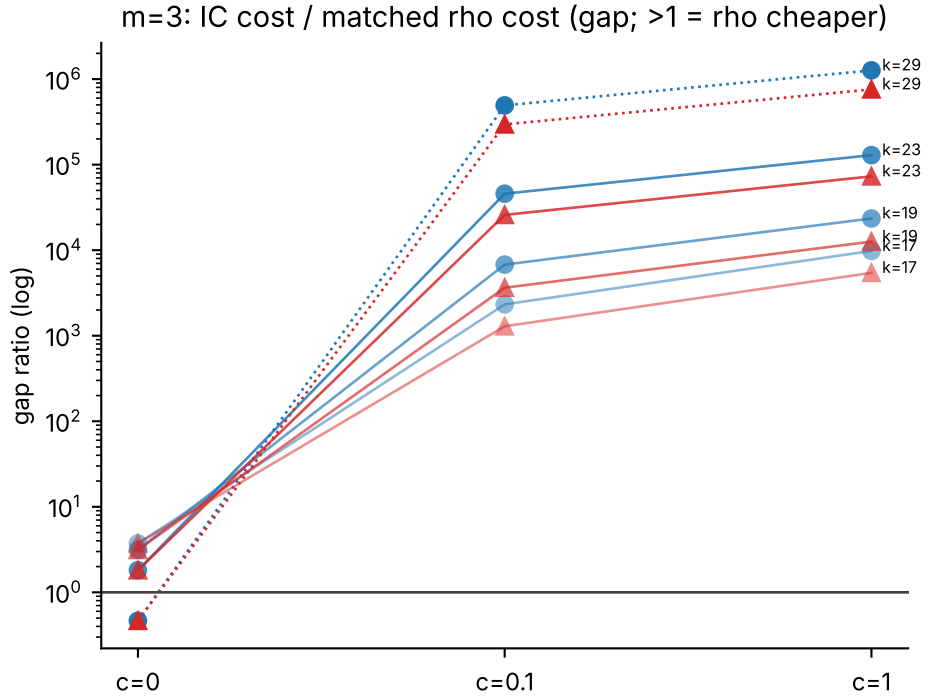

H1: non-liftable solutions per cell vs the independent-lift model
rate (log scale)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
coin-flip m=3: 0.875
coin-flip m=2: 0.75
95% upper bound on non-degenerate twist-coordinate rate (observed violations in bar label)
0 of 19522
0 of 20099
0 of 19691
0 of 9884
0 of 5918
0 of 7039
0 of 13260
0 of 2692
0 of 659
m=2 k=17
m=2 k=19
m=2 k=23
m=2 k=29
m=2 k=31
m=3 k=17
m=3 k=19
m=3 k=23
m=3 k=29


H2: twist fraction of factor-base columns, 170 curve-cells
N(0,1) (predicted)
density
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−4
−3
−2
−1
0
1
2
3
4
z = (f_twist - (1 - mu_E)) / sd, one per curve and cell (arm factor base)


solver-call ratio
measured mean U/F stage-1 calls
registered 2.0 / 4.0
call ratio
4.0
3.5
3.0
2.5
2.0
m=2 k=17
m=2 k=19
m=2 k=23
m=2 k=29
m=2 k=31
m=3 k=17
m=3 k=19
m=3 k=23
m=3 k=29


H3: cost ratio vs registered table
registered R (band +-10%)
measured R_hat (native ops, U/F)
cost ratio at c=0.1, 95% cluster CI
cost ratio U/F
1.8
1.6
1.4
1.2
1.0
m=2 k=17
m=2 k=19
m=2 k=23
m=2 k=29
m=2 k=31
m=3 k=17
m=3 k=19
m=3 k=23
m=3 k=29


m=2: IC cost / matched rho cost (gap; >1 = rho cheaper)
gap ratio (log)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
k=31
k=29
k=23
k=23
k=19
k=19
k=17
k=17
arm U
arm F
measured (d-determined)
extrapolated (registered N_need)
c=0
c=0.1
c=1


m=3: IC cost / matched rho cost (gap; >1 = rho cheaper)
gap ratio (log)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
k=29
k=29
k=23
k=23
k=19
k=19
k=17
k=17
c=0
c=0.1
c=1